Self-Supervised Spatial–Temporal Consistency for Source-Free Domain-Adaptive Segmentation in Remote Sensing Images

Zhihao Xi, Yu Meng, Yupeng Deng, Yuman Feng, Diyou Liu, and Jingbo Chen

***Abstract*—In conventional unsupervised domain adaptation (UDA), the model knowledge is transferred to the target domain with access to annotated source data, which is an unsuitable strategy for cross-domain scenarios involving data privacy and confidentiality. In this article, we focus on source-free domain adaptation (SFDA) for semantic segmentation tasks, which adapts source-trained models to unannotated target domains without relying on source data. Self-training paradigms dominate the existing approaches, but often suffer from significant performance degradation due to unreliable pseudolabels and knowledge forgetting. To address these challenges, we propose innovative self-supervised spatial–temporal consistency learning for source-free domain-adaptive segmentation (S^3T-SFDA). Specifically, to address pseudolabel ambiguity in various spatial contexts, a spatial multiview consistency (SMVC) mechanism is proposed to constrain the semantic consistency across different spatial views of the same image. To mitigate the knowledge forgetting problem caused by a lack of supervisory information, a temporal dynamic consistency (TDC) mechanism, which harnesses historical knowledge consistency to regularize the current model evolution direction, is proposed. Furthermore, to improve the category-discriminative representation capabilities under domain shifts, a spatial–temporal contrastive (STC) strategy is designed to promote the intrinsic semantic association of features belonging to different categories in the target domain. Extensive experiments were conducted on two domain-adaptive remote sensing (RS) segmentation benchmarks. The results demonstrate the flexibility of the proposed method when integrated into various advanced segmentation architectures, as well as its excellent generalization performance across different cross-domain RS scenarios.**

***Index Terms*—Remote sensing (RS) semantic segmentation, self-supervised learning, source-free domain adaptation (SFDA), spatial–temporal consistency.**

I. Introduction

REMOTE sensing (RS) image segmentation has achieved remarkable progress in recent years. However, its success largely hinges on two key assumptions: the availability of abundant labeled data and the consistency of data distributions between training and testing [1]. In real-world open environments, these assumptions rarely hold. The high cost of manual annotation severely limits labeled data, and variations in imaging sensors, geographic environment, and atmospheric radiation often lead to significant domain shifts, undermining model generalizability. To address this challenge, learning paradigms such as few-shot and semi-supervised learning have shown promise in reducing the reliance on annotations [2], [3], [4], [5]. Among them, unsupervised domain adaptation (UDA) has drawn increasing attention for transferring knowledge from the annotated source domain to the unannotated target domain [6], [7]. By mitigating domain shifts and alleviating annotation burdens, UDA has garnered significant attention in RS semantic segmentation tasks [8], [9], [10], [11].

In conventional UDA, most dominant methods rely on access to label-rich source data to transfer knowledge to the target domain effectively [12], [13], [14], [15]. Otherwise, the model is prone to training collapse because of the lack of supervision. However, this assumption is often unrealistic in RS scenarios. Data privacy and proprietary rights of annotated data have been the focus of attention for many data providers [16]. RS images in particular often involve sensitive geographic information and critical metadata, making them highly confidential and restricted from arbitrary retrieval, transfer, or release [17]. As a result, in practical scenarios such as cross-institution collaboration, commercial model delivery, or deployment of third-party models, direct access to source data is typically prohibited, and only the model itself can be shared for further adaptation. Under these constraints, the robust adaptation using only unlabeled target-domain samples presents a more practical and urgent challenge than conventional domain adaptation.

Source-free domain adaptation (SFDA) [16], [18], [19], [20] seeks to adapt well-trained or published models to target-domain scenarios without access to the source data. In this setting, the knowledge embedded within the model becomes the only available basis for supervision [21]. Recent research efforts have focused on leveraging self-training strategies, where pseudolabels generated by the model itself guide adaptation to the unlabeled target domain [20], [22], [23]. However, due to the absence of annotations and the significant distribution gap between domains, these pseudolabels often suffer from high uncertainty, making it difficult for the model to

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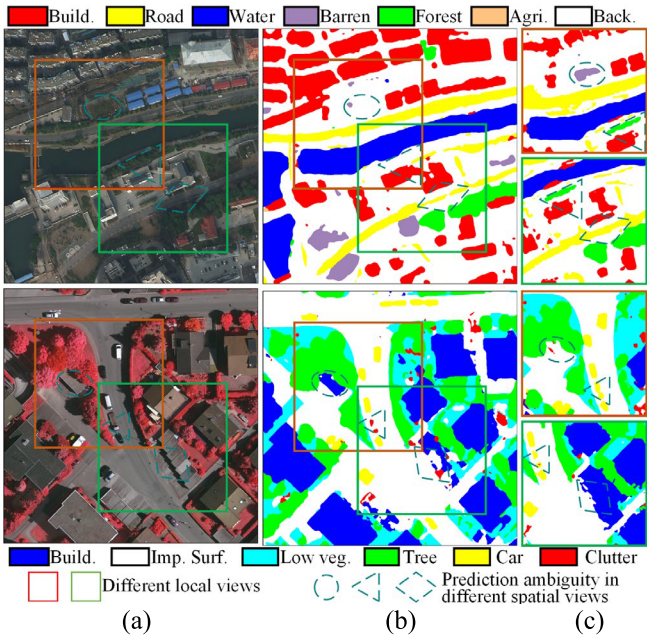


Fig. 1. Pseudolabel inconsistency across spatial views. (a) Target image. (b) Pseudolabels from the global views. (c) Pseudolabels from local views within the red and green boxes. Dashed regions highlight semantic inconsistencies arising from different spatial perspectives.

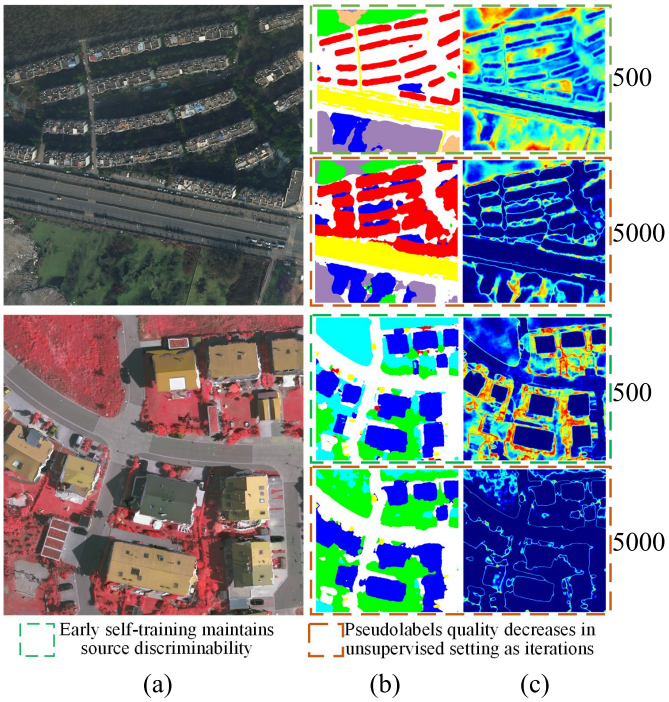


Fig. 2. Pseudolabels and predictive entropy from conventional SFDA at different iterations. The green dashed box indicates results at 500 iterations, and the red box at 5000 iterations. (a) Target images. (b) Pseudolabels at different iterations. (c) Visualized predictive entropy, where red regions indicate the higher uncertainty.

converge in a stable and effective direction. Numerous studies have regarded outputs with low-entropy or high-confidence predictions as reliable pseudolabels [24], [25], [26]. Although such strategies have shown promise in natural image domains, they face greater challenges in RS tasks, where blurry class boundaries, large variations in object scale, and complex spatial structures exacerbate prediction uncertainty and hinder adaptation performance.

Three critical factors for SFDA performance degradation in RS cross-domain scenarios require to be explored as follows.

1) *Pseudolabel Ambiguity:* RS images typically contain large feature scale variations and diverse spectral features, which render the pseudolabels generated by the model itself in a source-free setting prone to semantic inconsistencies. As illustrated in Fig. 1, the model produces inconsistent semantic predictions for the same region across different spatial perspectives, leading to instability during domain transfer.
2) *Knowledge Forgetting:* During unsupervised self-training, the model lacks constraints on early semantic representations, making it prone to drifting away from its initial discriminative features and becoming overly confident in its current predictions. As shown in Fig. 2, with continuous iterations, the current predictions exhibit noticeable semantic shifts compared with previous predictions. Despite higher confidence in the current predictions, the quality of the pseudolabels decreases, resulting in degraded adaptation performance.
3) *Decreased Discriminative Ability:* Without access to source data, the model fails to align the semantic distributions of the source and target domains, resulting in a category cognitive bias, which is more pronounced in RS images where geographic features are susceptible to confusion.

To improve the semantic consistency, adaptive stability, and discriminative ability of SFDA in source-free RS cross-domain scenarios, a novel self-supervised spatial–temporal consistency learning for source-free domain-adaptive segmentation (S^3T-SFDA) approach is proposed. Specifically, to address the semantic ambiguity caused by scale variation and contextual diversity in RS images, a spatial multiview consistency (SMVC) mechanism is introduced. It generates pseudolabels with different spatial contexts and performs fusion and alignment to constrain the predictive consistency in different spatial views, thus enhancing the stability and reliability of the pseudolabels. To alleviate the knowledge forgetting issue arising from the lack of source supervision, a temporal dynamic consistency (TDC) mechanism is proposed. It exploits the temporal consistency of model predictions during adaptation to estimate pseudolabel confidence dynamically. The consistency serves as the regularization weight for the pseudolabels, encouraging the model to retain historical structural knowledge and preventing overfitting to unreliable current predictions. Furthermore, to alleviate the cognitive biases caused by domain shifts, a category-level spatial–temporal contrastive (STC) strategy is designed, which enhances the discriminative ability for the semantic features by constructing category-level contrast learning across different temporal and spatial contexts.

The main contributions are summarized as follows.

1) A novel S^3T-SFDA approach for RS cross-domain scenarios is proposed, which mines the spatial–temporal

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contextual information in an unsupervised setting to improve adaptability without access to source data.

2) An SMVC mechanism, which improves the contextual consistency and reliability of pseudolabels through multiview fusion and prediction alignment, is proposed. In addition, a TDC mechanism is proposed to constrain the consistency between current and historical knowledge, preventing knowledge forgetting.
3) An STC strategy is designed to constrain category intrinsic semantic associations in a self-supervised manner through embedding across different temporal states and spatial views to improve discriminative representations.
4) Extensive experiments on RS cross-domain benchmarks demonstrate that S^{3}T-SFDA outperforms previous state-of-the-art (SOTA) methods and integrates seamlessly into mainstream segmentation architectures. The code is available at: https://github.com/yixinzs/S3T-SFDA

II. Related Works

A. UDA for RS Images

Conventional UDA adapts knowledge from label-rich source domains to label-scarce target domains, thus solving the model performance degradation problem caused by interdomain distribution discrepancies [27], [28]. Recent advances have achieved remarkable adaptation performance improvements in various cross-domain RS scenarios, such as cross-spectral [14], [29], cross-geographical [11], [30], [31], cross-sensor [32], [33], and multisource cross-domain settings [34], which can be technically characterized into two types: domain alignment [12], [35] and self-training [36], [37].

Domain alignment approaches leverage style transfer [38] and adversarial training [39] to bridge source–target feature gaps. Style transfer uses image transformation or GANs to adapt the visual style of source images to the target images, reducing the data distribution discrepancies at the input level. Adversarial training incorporates domain discriminators and adversarial losses to align feature distributions. Recent studies have introduced several variants to improve alignment effects, such as scale-aware alignment [33], fine-grained category alignment [40], and multilevel joint alignment [41]. However, domain alignment approaches rely on simultaneous access to both source and target data to align their distributions.

Self-training methods typically utilize pseudolabels to iteratively fine-tune the model on target data [42], [43], [44]. Their success hinges on pseudolabel quality, which can be improved through high-confidence thresholding [45], class balance strategies [46], and sample-adaptive filtering [47]. In addition, the self-training process tends to yield a massive amount of label noise due to domain shifts, and prototype-based distance measures and entropy-based uncertainty modeling have been proposed to rectify pseudolabel deviations [48], [49]. Nonetheless, most self-training approaches still rely on source data to guide adaptation, restricting their applicability when source data are unavailable.

Recently, consistency learning and feature constraints have emerged as promising solutions for improving adaptation stability and cross-domain generalization, such as prototype consistency, masked consistency, and feature distribution consistency. PCEL aligns multiple prototypes to strengthen cross-domain category relations and incorporates mask consistency for contextual robustness [48]. MGLP enforces logical prototype consistency at both coarse- and fine-grained levels within the same class space [50]. DLC exploits higher order neighborhood consistency to model the target noise distribution. MS-CADA integrates multisource knowledge via cross-domain mixed consistency, exploiting the complementarity of multiple source experts [34]. Consistency constraints have also attracted attention for cross-scene generalization in foundation models. HyperSIGMA constrains both spectral and spatial features across multiple sources to extract general, scene-independent knowledge [51]. SpectralEarth introduces spectral adapters to accommodate band-specific distributions for task and sensor generalization [52]. HyperSL aligns spectral features from multiple sources to improve cross-scene transferability [53].

Despite these advances, conventional UDA methods remain dependent on the labeled source data. Without source supervision, models face severe performance degradation issues, reducing their practicality in collaborative or deployment scenarios in which source data cannot be shared, and thus limiting their application in data-protected cross-domain settings.

B. Source-Free Domain Adaptation

In challenging cross-domain scenarios where source data are inaccessible, SFDA aims to adapt well-trained models to the target domain while utilizing only the unannotated target data in an unsupervised manner. With respect to segmentation tasks, self-training has evolved as the predominant approach for solving the SFDA problem [21], [22], [54]. Consequently, the stability and reliability of pseudolabels have become key research focuses concerning many SFDA methods, which widely employ confidence-based, consistency-based, and prototype-based learning strategies to obtain reliable pseudolabels [55], [56]. C-SFDA utilizes a curriculum-style self-training strategy that prioritizes training on high confidence, easy-to-learn samples before gradually propagating to unreliable, hard-to-learn samples [24]. AUGCO leverages the prediction consistency across different augmented views to identify reliable pixel predictions [57]. IAPC employs an importance-aware prediction reliability measurement mechanism to correct for target-domain prediction biases [56]. To mitigate the confirmation bias, SND observes that current predictions exhibit greater consistency with the initial model showing more stability, thus identifying stable and unstable target samples [20]. Nevertheless, the pseudolabel learning is inherently sensitive to data shifts, which are further amplified due to the absence of source data.

Furthermore, unconstrained self-training can easily lose the self-evolution direction, resulting in catastrophic performance degradation. DT-ST uses a dynamic teacher update (DTU) mechanism to evaluate the evolving state of the model [21]. To avoid parameter overfitting, Tent optimizes only the batch-norm parameters of the adapted model and prevents task drifts by freezing all other parameters [26]. Despite these advances, existing SFDA methods are confined to single view,

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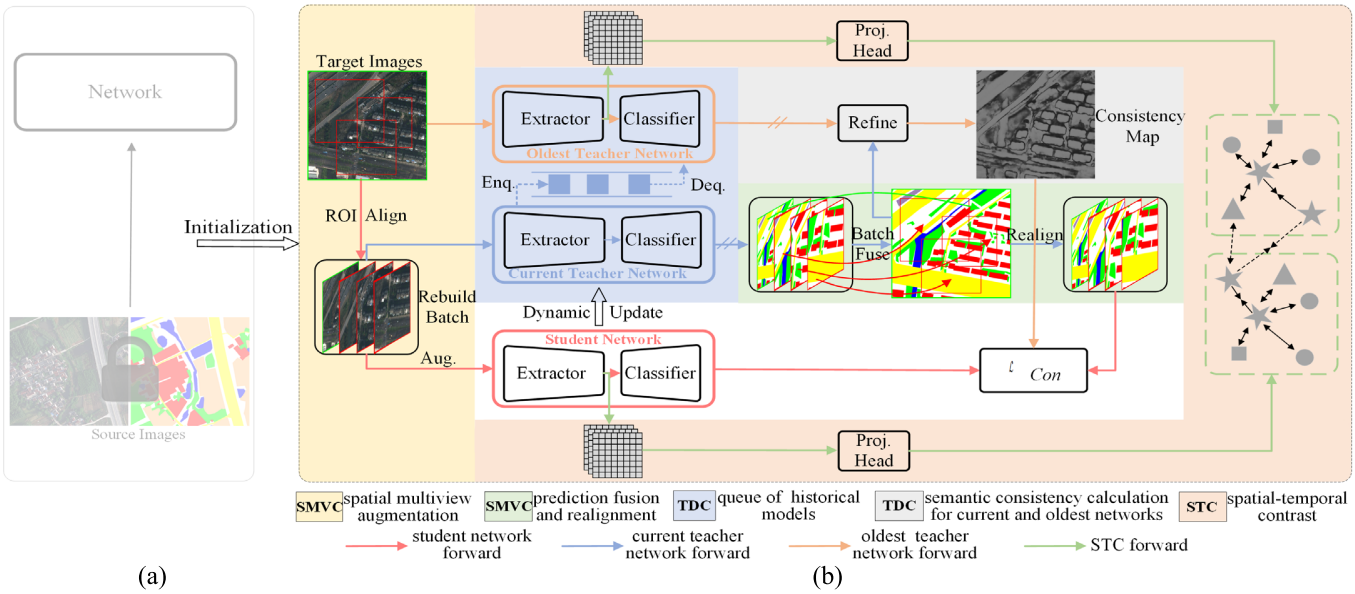


Fig. 3. Overview of the proposed S^{3}T-SFDA architecture. (a) Segmentation model is first trained on the source domain. (b) In the SFDA stage, only unannotated target images are used. Each target image is reconstructed into multiple spatial views via ROIAlign and then fed to the student and current teacher networks with different augmentations. Teacher predictions are fused and realigned to produce spatially consistent pseudolabels, forming the SMVC process. To improve stability, a historical teacher queue is maintained. The original images are fed to the oldest teacher, and its outputs are compared with those from the current teacher to refine pseudolabel reliability. A weighting map is generated to guide the consistency loss and regularize the evolution direction, a process termed TDC. Finally, features from the student and the oldest teacher are projected into uniform dense embeddings via the projection head. The STC loss then aligns semantic features across spatial and temporal domains, encouraging the model to learn the intrinsic semantic associations of different categories.

the static history representation for assessing the reliability of pseudolabels, ignoring the spatial semantic drift and dynamic noise accumulation during self-evolution. In contrast, our approach involves a dual-constraint strategy of multiview consistency and dynamic history consistency to mitigate pseudolabel ambiguity and semantic drift from both spatial and temporal perspectives.

C. Contrastive Learning

As a classic self-supervised representation learning framework, contrastive learning generally involves discriminative representations in which similar instances are pulled closer and different instances are pushed away to achieve intrinsic semantic consistency [58]. Early work on contrastive learning focused primarily on designing proxy tasks, introducing InfoNCE loss to encourage models to acquire more robust visual representations that could be better transferred to downstream tasks [58], [59], [60]. Later studies validated the effectiveness of contrastive learning in UDA [61] and semantic segmentation [62], where it typically serves as an auxiliary task for mining image-intrinsic contextual information. ProCA uses a prototype-based classwise contrastive strategy to achieve domain-adaptive semantic segmentation [63]. PiPa further employs a multigrained intraimage contrastive approach to extract domain-invariant contextual representations [64]. However, the effectiveness of these methods relies on simultaneous access to both source and target data. CMA introduces a domainwise contrastive approach under the restriction of prohibiting source data; it still requires additional reference images, which increases the data acquisition demands [65]. HCL [66] designs a straightforward historical contrastive learning mechanism to obtain instance-level discriminative representations. In contrast to prior works that emphasize contrastive learning across augmented views [62], [64], our method introduces contrastive constraints between diverse spatial contexts and temporal representations to extract more stable and discriminative semantic features for SFDA.

III. Methodology

This section details the proposed S^{3}T-SFDA approach. It leverages spatial–temporal consistency and contrastive learning to adapt a source-trained model to the target domain, which suffers from distributional discrepancies, using only unannotated data. As illustrated in Fig. 3, the S^{3}T-SFDA consists of three core components: SMVC, TDC, and STC. The SMVC mechanism generates unified and stable pseudolabels by fusing and aligning multiview predictions, mitigating the semantic ambiguity across spatial contexts (see Section III-B). The TDC mechanism leverages historical knowledge to guide the self-evolution of the current model, alleviating the knowledge forgetting issue caused by the absence of supervision (see Section III-C). The STC strategy constructs contrastive losses across spatial views and temporal states to enhance category-discriminative representations, thereby improving the robustness and generalization to the target domain (see Section III-D). Furthermore, our method is designed to support advanced transformer-based architectures and can seamlessly integrate with existing SFDA approaches.

A. Self-Training for SFDA

In general, UDA, $\mathcal{D}_S = \{(x_s, y_s)|x_s \in \mathbb{R}^{H\times W\times 3}, y_s \in \mathbb{R}^{H\times W}\}$, represents the annotated source data, $f_{\theta_s} : x_s \rightarrow y_s$ denotes

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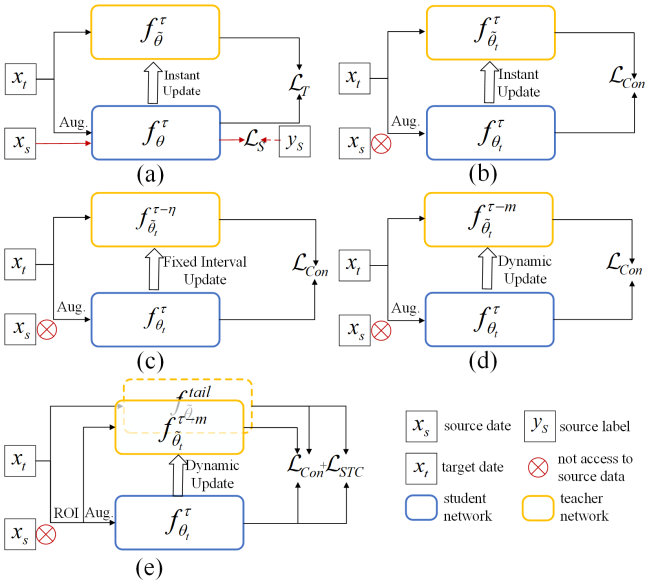


Fig. 4. Different domain adaptation schemes and distinct update paradigms for teacher networks. (a) Conventional UDA. (b) Instant momentum updating for SFDA. (c) Fixed-interval momentum updating for SFDA. (d) DTU for SFDA. (e) Ours.

the model trained on the source domain, and θ_s represents the model parameters. $\mathcal{D}_T = \{(x_t)|x_t \in \mathbb{R}^{H\times W\times 3}\}$ is defined as the unannotated target data, where the source and target domains share the same semantic categories $\mathcal{C} = \{1, 2, \ldots, K\}$. UDA inherently requires accessing both the source domain $\mathcal{D}_S$ and the target domain $\mathcal{D}_T$ to adapt the source-trained knowledge to the target domain. However, SFDA involves adapting the well-trained model f_{θ_s} to the target domain when the source data $\mathcal{D}_S$ are inaccessible, thus posing more challenges in this task.

We adopt online self-training (ONST) to construct SFDA for semantic segmentation, ensuring efficient and flexible dynamic adaptation throughout the iterative process. Fig. 4(a) illustrates the conventional online UDA paradigm, where both source and target data are accessible during training. The paradigm consists of two components, a teacher network $f_{\tilde{\theta}_t}$ and a student network f_{θ_t}, which share the same structure. The teacher network is employed to generate pseudolabels, while the student network is involved in backpropagation training, and the parameters of the teacher network are updated via an exponential moving average (EMA), i.e., $\tilde{\theta}_t = (1-\lambda)\theta_t + \tilde{\theta}_t$, where λ is the momentum coefficient. In a typical UDA task, the training loss can be formulated as follows:

$$\mathcal{L}_{\mathrm{DA}} = \mathcal{L}_S\left(f_{\theta_s}(x_s), y_s\right) + \mathcal{L}_T\left(f_{\theta_t}(x_t), f_{\tilde{\theta}_t}(x_t)\right) \tag{1}$$

where $\mathcal{L}_S$ represents the supervised loss induced on the source domain, which is used to capture source semantic knowledge, and $\mathcal{L}_T$ denotes the unsupervised loss to adapt knowledge from the source domain to the target domain. Both $\mathcal{L}_S$ and $\mathcal{L}_T$ are typically pixelwise cross-entropy losses. However, as shown in Fig. 4(b), when source data are protected and inaccessible, the source-trained model f_{θ_s} and its parameters θ_s become the only effectively available knowledge. Therefore, θ_s is generally taken directly as the initialization parameter for the student network f_{θ_t} and the teacher network $f_{\tilde{\theta}_t}$ at the beginning of the training process, i.e., $\theta_s = \theta_t = \tilde{\theta}_t$. The training loss of the SFDA is reformulated as follows:

$$\mathcal{L}_{\mathrm{DA}} = \mathcal{L}_T\left(f_{\theta_t}(x_t), f_{\tilde{\theta}_t}(x_t)\right). \tag{2}$$

Since the frequent momentum-based update strategy easily leads to performance degradation, the fixed-interval momentum-based update mechanism provides a more stable alternative [55]. As illustrated in Fig. 4(c), in contrast to conventional teacher networks that update instantaneously through a student network, those equipped with appropriate momentum update intervals [21], [55] have been demonstrated to alleviate model degradation problems. The fixed-interval momentum-based update process can be rewritten as follows:

$$\tilde{\theta}_t^{\tau} = (1-\lambda)\,\theta_t^{\tau} + \lambda\tilde{\theta}_t^{\tau-\eta} \tag{3}$$

where τ represents the current iteration and η denotes the update interval for the teacher network, with a larger η implying slower updates.

However, due to the absence of effective supervision information from the source domain, the model is prone to losing its self-evolution direction. Inspired by the design of DTU [21], as shown in Fig. 4(d), the predictive entropy is used to measure whether the historical samples have been optimized, thus determining the update interval for the teacher network. This means that the teacher network can be selectively updated based on the evolving performance of the student network. In this way, the update interval varies dynamically over a certain range with increasing iterations, and this dynamic interval is set to m. The training process of SFDA can be further formulated as follows:

$$\mathcal{L}_{\mathrm{DA}} = \mathcal{L}_T\left(f_{\theta_t}^{\tau}(x_t), f_{\tilde{\theta}_t}^{\tau-m}(x_t)\right). \tag{4}$$

Although DTU provides an evolutionary direction for SFDA by leveraging feedback from a small set of historical samples, it lacks a direct connection between current inputs and their predictions. We assume that performance degradation is caused primarily by the unstable variations in pseudolabels generated from current inputs. Thus, enhancing the quality and stability of these current pseudolabels plays a more critical role in optimization. Furthermore, as mentioned earlier, in the absence of supervisory signals, the self-training is prone to overconfidence in the current predictions and deviates from previously learned knowledge, resulting in low-entropy yet unreliable pseudolabels [25].

In summary, to improve the stability and adaptability of SFDA, three key directions can be pursued to further improve the generalization: 1) generating stable pseudolabels via spatial multiview augmentation and prediction fusion; 2) regularizing the evolution direction using knowledge from historical states, which enables the model to self-correct noisy pseudolabels and avoid overconfidence in recent predictions; and 3) introducing a pixel-level, source-free self-supervised paradigm that leverages historical and multiview features to encourage the discriminative representation learning with strong intraclass compactness and interclass separability.

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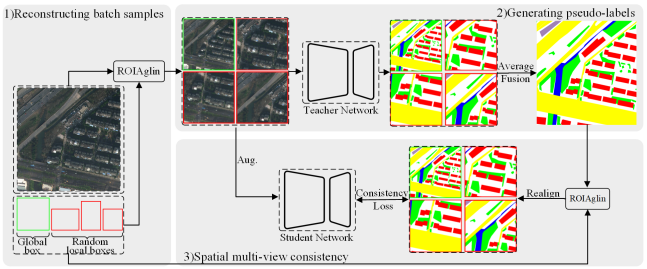


Fig. 5. Illustration of the proposed SMVC mechanism.

B. Spatial Multiview Consistency

Owing to the variations in spatial contextual information, the model is prone to presenting prediction ambiguities for the same image region under different local spatial perspectives, resulting in unstable pseudolabels that hinder effective adaptation in unsupervised settings. Existing optimization methods are not powerful enough to efficiently constrain semantic consistency across contexts. To address this issue, we propose an SMVC mechanism under source-free conditions, as illustrated in Fig. 5. It enforces semantic fusion and alignment across multiple spatial views for corresponding pixel regions, thereby improving its adaptation capability.

For an original image with a size of $H \times W$, a constant size is maintained by following the conventional augmentation strategy. Subsequently, $N-1$ region boxes $\{b_c^l\}_{n=1}^{N-1}$ are randomly generated with different local views and overlap conditions. These region boxes are fed into the ROIAlign module [67] along with the bounding boxes of the original image to be reconstructed into a new batch of samples, yielding N images $\{x_t^l\}_{l=1}^{N}$ with different spatial views of size $H/2 \times W/2$. Note that the overlaps between different spatial-view images range between 10% and 50%, such that the spatial perspectives and contextual information vary across different images in terms of the overlapping pixel regions. Formally, this is expressed as follows:

$$x_t^l = \text{ROIAlign}\left(x_t, b_c^l\right). \tag{5}$$

The reconstructed samples are fed into the student and teacher networks. Following the principle of weak-to-strong augmentation consistency learning [4], the student network introduces moderate input perturbations through stronger augmentations, while preserving the original spatial information of the augmented images. Simultaneously, the teacher network is designed to generate high-quality pseudolabels, which serve as reliable supervisory information for optimizing the student network. Specifically, its predictions are constrained to share consistent semantics for the same spatial region across different spatial views, ensuring that the generated pseudolabels are both reliable and of high quality. To achieve this, a two-step strategy is designed: 1) average prediction fusion and 2) label realignment. First, the predictions are restored to the original resolution via the inverse ROIAlign process. Then, for overlapping regions, the predictions are fused via an averaging operation to ensure semantic consistency

$$\hat{y}_T^{\tau-m} = \frac{1}{N}\sum_{l=1}^{N} \mathcal{M}\left(f_{\tilde{\theta}_t}^{\tau-m}\left(x_t^l\right)\right) \tag{6}$$

where x_t^l represents the input acquired from different spatial perspectives, $\mathcal{M}$ denotes the mask function derived from the region boxes, N refers to the number of predictions to be fused, and $\hat{y}_t^{\tau-m}$ represents the global prediction after performing fusion, with a resolution of $H \times W$. After the prediction $\hat{y}_t^{\tau-m}$ is obtained, ROIAlign is utilized to conduct realignment again with the region boxes $\{b_c^l\}_{n=1}^{N}$, ensuring that the same pixel regions across different spatial-view images yield consistent semantic content. This fusion process functions similar to model ensembling, enabling the generation of more reliable pseudolabels. The aligned pseudolabels are then used as supervisory signals during self-training to train the student network, achieving SMVC learning for SFDA.

C. Temporal Dynamic Consistency

As model parameters are continuously updated during adaptation, source-free self-training can easily suffer from knowledge forgetting, where earlier learned representations are gradually lost, and overconfidence is placed on current predictions. Consequently, many adaptation methods that tend to treat low-entropy or high-confidence current predictions as pseudolabels are prone to label bias and misguided optimization [24], [36], [47]. To this end, we propose a TDC mechanism, which evaluates the semantic consistency of predictions for the same image across different iterations within a dynamic temporal window. The pseudolabel losses are then reweighted based on this consistency to regularize the training direction and enhance the stability of adaptation.

The core of this mechanism lies in maintaining a queue of M historical models during self-training, thus providing the ability to reutilize past iterations to evaluate the semantic consistency between the outputs of the current and historical models. The queue is incrementally refreshed through the momentum update process, with the new teacher model placed in the queue and the outdated model removed after every K momentum update. This ensures that the models contained in the queue remain within a controllable iteration range, as outdated knowledge tends to lose semantic consistency with the evolving current model. In particular, the historical model queue is designed based on a DTU mechanism [21], such that although the queue size remains fixed, the iteration intervals between models are dynamically varied within a margin, with such variations encouraging SFDA to be adaptable to different iterative cases.

In the model queue, we only calculate the consistency between the current teacher network $f_{\tilde{\theta}_t}$ and the oldest teacher network $f_{\tilde{\theta}_t}^{\text{tail}}$ at the tail of the queue. As previously mentioned, the inputs of the current teacher network are organized into different spatial perspectives, whereas the inputs of the historical teacher network correspond to original images that have not been augmented. However, both have output resolutions of $H \times W$ so that the semantic consistency between the current generated pseudolabels and those of the historical model can be calculated as follows:

$$w^{\tau} = 1 - \text{sigmoid}\left(\text{KL}\left(f_{\tilde{\theta}_t}^{\tau}\left(x_t^{\tau}\right), f_{\tilde{\theta}_t}^{\text{tail}}\left(x_t^{\tau}\right)\right)\right) \tag{7}$$

where w^{τ} represents the consistency weighting map evaluated based on the current and oldest teacher networks, and the

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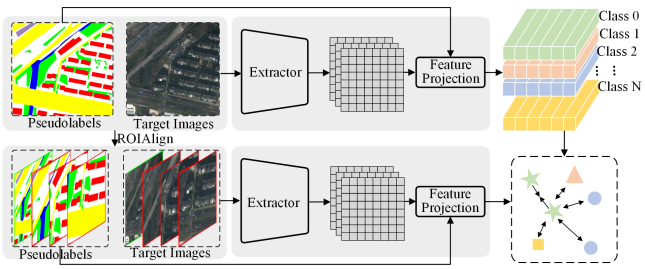


Fig. 6. Illustration of the proposed STC strategy.

sigmoid function ensures that $w^{\tau} \in [0, 1]$. KL denotes the Kullback–Leibler divergence calculation function. We regularize the evolution direction by reassigning weights for the pseudolabels to determine the training consistency loss

$$\mathcal{L}_{\text{Con}} = \sum_{i,j}^{H,W} w^{\tau,i,j} \hat{y}_t^{\tau-m,i,j} \log\left(p\left(f_{\theta_t}^{\tau}\left(x_t^{\tau}\right)\right)\right) \tag{8}$$

where $\hat{y}_t^{\tau-m,i,j}$ represents the pseudolabels generated by the current teacher network, which are obtained via (6). A larger weight w^{τ} indicates that the prediction of the corresponding sample behaves more stably during different periods of evolving model, suggesting greater pseudolabel reliability and a stronger contribution to training. In contrast, samples with unstable predictions are assigned weights that tend to 0 during the evolution process. In this way, this process facilitates the self-evolution of the model and prevents its optimization procedure from suffering perturbations caused by unstable label noise.

D. Spatial–Temporal Contrast

The semantic segmentation is essentially a pixel grouping problem, in which different semantic categories are assigned to pixel groups based on their contextual consistency. Therefore, guiding the segmentation network to learn intrinsic contextual cues and discriminative category representations is critical. As shown in Fig. 6, we encourage the network to learn the intrinsic semantic associations among different categories in a self-supervised manner, using only unlabeled target images. Distinct from previous contrast learning in UDA [63], [64], we propose the STC strategy to compensate for the absence of source data, which often leads to weakened discriminability. By contrasting features across multiple spatial views and temporal states, STC promotes the learning of more robust and transferable category-discriminative representations, thereby improving generalizability to the target domain.

Given a target image x_t, multiview spatial transformations are applied to reconstruct batch samples, which are then input into the student network. The feature extractor of the student network generates embedded features F_t^{τ}, and the current teacher network produces corresponding pseudolabels $\hat{y}_t^{\tau-m}$. Simultaneously, we obtain the historical embedded features F_t^{tail} of the target image from the oldest teacher network $f_{\theta_T}^{\text{tail}}$. Anchors and positive samples generally represent similar semantics under different visual conditions. Hence, we treat the encoded embeddings of trainable student networks as anchor samples, and the encoded embeddings projected by pseudolabels belonging to the same semantic categories are regarded as positive samples. Conversely, embeddings belonging to other semantic categories serve as negative samples. The category-level STC loss is mathematically expressed as follows:

$$\mathcal{L}_{\text{STC}} = \sum_{C_i=C_j} \log \frac{\exp\left(f_i \cdot f_j/\varepsilon\right)}{\sum_{k=1}^{N_p} \exp\left(f_i \cdot f_k/\varepsilon\right)} \tag{9}$$

where f_i represents the embeddings mapped through the projection head h:, i and j denote the ith and jth embedding elements, respectively; N_p indicates the number of embeddings; ε is the temperature parameter. Under the guidance of category-level STC loss, the model explicitly pulls the pixel embeddings of the same semantic category closer together while pushing away the embeddings of different categories. This encourages the model to learn category-discriminative representations for the target domain in an unsupervised setting, thereby effectively generalizing the knowledge to the target domain.

To reduce the computational overhead, we leverage an independent projection head h consisting of a two-layer MLP to further map the dense embedded features F_t^{τ} and F_t^{tail} to the 128-D embedding space. Notably, unlike the conventional approach, which uses different augmented data to conduct contrastive learning, we extend the positive samples not only from the embeddings derived from trainable student networks with different spatial perspectives but also from the embeddings of the historical models. To optimize the embedding space of the segmentation network through the multilevel STC strategy, the model is prevented from losing its semantic discrimination ability with iterations because of the absence of supervisory information.

E. Total Training Loss

Similar to previous efforts that employed ONST for UDA tasks [22], [42], although the S^{3}T-SFDA can improve the reliability and stability of the pseudolabel generation process, it still faces the problems of category bias and hard samples in the target domain, causing the performance degradation. To address this, we follow the instance-adaptive pseudolabel selection strategy [47] to reduce the data redundancy and increase the diversity of the pseudolabels. Specifically, the pseudolabels predicted by the teacher network combined with the SMVC and TDC mechanisms serve as reliable supervision signals, and the multicategorical cross-entropy loss is further used to construct $\mathcal{L}_{\text{Con}}$. In addition, to facilitate the model in diversifying its predictions and prevent the model from assigning all the output probabilities to the most probable category, we introduce an information entropy loss $\mathcal{L}_E$

$$\mathcal{L}_E = \frac{1}{H \times W} \sum_{i,j}^{H,W} \sum_{c}^{C} p\left(\hat{y}_t^{i,j,c}\right) \log\left(p\left(\hat{y}_t^{i,j,c}\right)\right). \tag{10}$$

Ultimately, with the joint optimization of the information entropy loss [26], STC loss, and consistency loss constructed

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Algorithm 1 Training Process of S^{3}T-SFDA

Inputs: student network f_{θ_t}, current teacher network $f_{\tilde{\theta}_t}^{\tau-m}$ with a dynamic update interval m, oldest teacher network $f_{\tilde{\theta}_t}^{\text{tail}}$, projection head h, source-trained weights θ_s, target images $\mathcal{D}_T = \{x_t^{(i)}\}_{i=1}^{N_t}$, maximum number of iterations Γ

Initialize the network weights $\tilde{\theta}_t = \theta_t = \theta_s$
for $i = 1$ to Γ do
  Randomly select samples x_t from $\mathcal{D}_T$ and the overlapping region boxes $\{b_c^l\}_{l=1}^N$, followed by reconstructing batch images $\{x_t^l\}_{l=1}^N$ using Eq. (5)
  Obtain predictions $\hat{y}_t^\tau$ and embeddings F_t^τ from trainable student networks: $\hat{y}_t^\tau, F_t^\tau \leftarrow f_{\tilde{\theta}_t}^\tau(x_t^\tau)$
  Generate predictions $\hat{y}_t^{\tau-m}$ with Eq. (6) and realign them to construct multiview consistent pseudolabels $\{y_t^l\}_{l=1}^N$: $y_t^l = \text{ROIAlign}(\arg\max(\hat{y}_t^{\tau-m}), b_c^l)$
  Obtain predictions $\hat{y}_t^{\text{tail}}$ and embeddings F_t^{tail} from oldest teacher network: $\hat{y}_t^{\text{tail}}, F_t^{\text{tail}} \leftarrow f_{\tilde{\theta}_t}^{\text{tail}}(x_t^\tau)$
  Calculate the consistency weighting map w^τ using $\hat{y}_t^{\tau-m}$ and $\hat{y}_t^{\text{tail}}$ via Eq. (7)
  Calculate the consistency loss $\mathcal{L}_{\text{Con}}$ using $\hat{y}_t^\tau, y_t^l$ and w^τ via Eq. (8)
  Calculate the STC loss $\mathcal{L}_{\text{STC}}$ using F_t^τ and F_t^{tail} via Eq. (9)
  Optimize the student network f_{θ_t} through the overall loss $\mathcal{L}_{\text{total}}$ of Eq. (10)
  Update the current teacher network $f_{\tilde{\theta}_t}^{\tau-m}$ and model queue via DTU [21] and TDC
end for
Return adapted segmentation network f_{θ_t}

by SMVC and TDC, the overall loss for training SFDA in the presence of only unannotated target data is as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Con}} + \alpha\mathcal{L}_{\text{STC}} + \beta\mathcal{L}_E \tag{11}$$

where α and β are used to determine the relative contribution to the training loss during the gradient update process, with default values set to 0.1 and 0.5, respectively. Simultaneously, we summarize the training procedure of the S^{3}T-SFDA in Algorithm 1.

IV. Experimental Results and Analyses

A. Dataset Description

To validate the adaptability of the proposed method to cross-domain tasks that do not provide access to the source data, three RS benchmark datasets, ISPRS Potsdam, ISPRS Vaihingen, and LoveDA, are used to conduct the experiments.

Potsdam Dataset: The Potsdam dataset provides 38 high-resolution ortho-6000 × 6000-pixel RS images with a spatial resolution of 5 cm. Each image provides four bands, i.e., red, green, blue, and near-infrared, where the IR–R–G imaging mode is primarily used in the subsequent experiments. The dataset has been manually annotated into six semantic categories: impervious surfaces, buildings, trees, cars, low vegetation, and clutter/background. Following the data splitting settings described in [14], [33], and [72], 17 images are used as the training set, seven images are used as the validation set, and the remaining 14 images are used as the testing set.

Vaihingen Dataset: The Vaihingen dataset contains 33 high-resolution true orthophotos with sizes of nearly 2000 × 2000 pixels and a spatial resolution of 9 cm. Akin to the Potsdam dataset, it includes the same six semantic categories but is available only in the IR–R–G imaging mode. Specifically, 11 images are selected as the training set, five images are selected as the validation data, and the remaining 17 officially assigned images are used as the testing set [14], [33], [72]. To facilitate training, the Vaihingen and Potsdam datasets are cropped to 512 × 512 pixels.

LoveDA Dataset: The LoveDA dataset comprises 5987 0.3-m high-resolution images with sizes of 1024 × 1024. It provides urban and rural scene images acquired from three regions, i.e., Nanjing, Changzhou, and Wuhan, which are annotated into seven semantic categories: building, road, water, barren, forest, agriculture, and background. The urban scene of this dataset contains 1156 training images, 677 validation images, and 820 testing images, whereas the rural scene contains 2358 images, of which 1366 images are used for training, and 976 are employed as testing data [31], [73].

B. Evaluation Metrics

Following the setup employed in related work [70], we use two common evaluation metrics, the intersection over union (IoU) and $F1$ score, to validate the performance of the proposed method. The IoU and $F1$ score can be calculated as follows:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{12}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{13}$$

where TP, FP, and FN can be calculated via the confusion matrix, and Precision and Recall are formulated as Precision = TP/(TP + FP) and Recall = TP/(TP + FN), respectively.

C. Implementation Details

We conduct experiments based on convolutional neural networks (CNNs) and transformer architectures, both of which are implemented with the PyTorch framework. For the CNN-based approach, we use DeepLabV2 [74] with ResNet101 [75] as the backbone, which serves as the baseline for domain adaptation tasks. During training, we employ the SGD optimizer with a momentum of 0.9 and a weight decay rate of 10^{-6}. The initial learning rate is set to 2.5×10^{-4}, and it decays according to a polynomial strategy with a power of 0.9. For the transformer-based approach, we adopt SegFormer [76] with the MiT-B2 backbone as the segmentation baseline and optimize it through the AdamW [77] optimizer with a momentum parameter of 0.9 and a weight rate decay of 10^{-2}. The learning rates are initialized to 6×10^{-5} and 6×10^{-4} for encoder and decoder, respectively, and they linearly decay throughout the training process.

All the experiments are conducted on an RTX 3090 GPU with a batch size of 4, and the models are trained for 20k

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TABLE I

QUANTITATIVE COMPARISON RESULTS FOR THE POTSDAM→VAIHINGEN TASK. SF INDICATES WHETHER DOMAIN ADAPTATION WAS PERFORMED UNDER SOURCE-FREE CONDITIONS. PERFORMANCE VALIDATED ON THE VAIHINGEN TESTING SET

Methods	SF	Imp. Surf.		Building		Low veg.		Tree		Car		Clutter		mIoU	
		F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	mFscore	mIoU
DeeplabV2	-	62.26	45.20	75.78	61.00	52.34	35.45	75.62	60.80	53.12	36.17	16.04	8.72	55.86	41.22
SegFormer	-	77.29	62.99	85.08	74.03	63.38	46.39	79.84	66.44	65.63	48.84	24.31	13.84	65.92	52.09
scaleDA[33]	✗	-	55.22	-	64.46	-	31.34	-	50.40	-	39.86	-	-	-	47.66
Zhang et al.[68]	✗	80.13	67.74	86.65	76.75	64.16	47.02	71.90	55.03	61.94	44.90	31.34	20.71	66.02	52.03
JDAF[69]	✗	81.49	68.76	87.13	77.19	64.30	47.39	73.72	58.38	59.90	42.76	81.49	68.76	70.38	55.52
DAFormer[13]	✗	86.72	76.55	91.04	83.56	67.71	51.19	80.26	67.03	68.82	52.46	62.15	45.09	76.12	62.65
ST-DASegNet[70]	✗	85.36	74.43	92.03	85.23	65.37	48.57	80.49	67.36	60.49	43.38	80.28	67.03	77.34	64.33
PFM-JONet[71]	✗	83.83	72.16	91.19	83.81	70.46	54.39	81.82	69.23	69.00	52.68	81.57	68.88	79.65	66.86
HCL[66]	✔	76.24	61.60	85.13	74.11	58.30	41.14	68.38	51.95	64.38	47.47	3.74	1.91	59.36	46.37
IAPC[56]	✔	74.61	59.51	84.12	72.59	61.86	44.78	71.21	55.30	62.62	45.58	27.16	15.72	63.60	48.91
DT-ST[21]	✔	83.56	71.76	85.28	74.34	61.87	44.79	72.21	60.58	54.30	39.16	34.37	20.96	65.27	51.93
SND[20]	✔	82.06	69.58	85.72	75.02	64.46	47.56	74.31	60.71	55.37	39.24	28.45	17.64	65.06	51.62
S^3T-SFDA(DLv2)	✔	81.23	68.40	87.59	77.92	**68.54**	**52.14**	75.97	61.25	54.66	37.61	41.15	25.91	68.19	53.87
S^3T-SFDA(SegF.)	✔	**88.27**	**79.00**	**92.37**	**85.83**	67.1	50.49	**79.59**	**66.10**	**71.69**	**55.87**	**64.04**	**47.10**	**77.18**	**64.06**

iterations in all the SFDA tasks. S^3T-SFDA obeys weak-to-strong augmentation consistency training [4], [45], where the teacher network is weakly augmented via random flipping and cropping, and the student network is further strongly augmented by random scaling, color jittering, and random cropping. In the SMVC process, we randomly sample $N = 4$ ROIs from each original image. The bounding box of each ROI is randomly selected within the range of $(H/2, W/2)$–(H, W), ensuring moderate overlap between sampled regions, while allowing the aspect ratio to vary within (0.8, 1.2). The training process involves two steps: implementing supervised training using the annotated source data to obtain a source-trained weight and then performing domain adaptation using only the well-trained weight and the unannotated target data to improve the generalization of the model in the target domain.

D. Experimental Results

We compare the proposed method with the SFDA [18], [20], [21], [56], [66] and conventional UDA methods [13], [70], [71], [78] in semantic segmentation research. Compared with the source-trained baseline, our approach achieves promising performance improvements in the target domain through SFDA, realizing SOTA performance in various RS cross-domain scenarios. Notably, S^3T-SFDA, which does not rely on source data, achieves performance comparable with those of conventional UDA methods.

1) Potsdam→Vaihingen: We conduct experiments on the Potsdam→Vaihingen cross-domain task in source-free settings. The two data scenarios have obvious discrepancies in image resolution, spatial scale of features, and building landscape, resulting in significant cross-domain distribution shifts. As shown in Table I, compared with the CNN-based source-trained models, the proposed method achieves an mIoU of 53.87% and an mFscore of 68.19%, representing improvements of 12.65% and 12.33%, respectively. Taking SegFormer with the transformer architecture as the source-trained baseline, its mIoU performance is 52.09%, and its mFscore performance is 65.92%, whereas our S^3T-SFDA further surpasses the average adaptability to 64.06% and 77.18%, respectively.

Compared with existing SFDA methods that employ the same source-trained weights, our method achieves superior adaptation performance, with 2.92% and 1.94% higher mIoU and mFscore than the SOTA method of DT-ST. Notably, under the strict source-free setting, S^3T-SFDA achieves target-domain performance comparable to mainstream UDA methods that access both source and target data, with only an approximately 2.8% drop in mIoU. This highlights the effectiveness of our method in leveraging the intrinsic domain knowledge embedded in the source model. In addition, the per-class IoU results demonstrate strong adaptability to categories such as buildings, trees, and low vegetation. As illustrated in Fig. 7(a), S^3T-SFDA produces more accurate and structurally coherent segmentation across all semantic classes, particularly for buildings, validating the effectiveness of the proposed SMVC module in enhancing spatial consistency and adapting to multiscale objects.

2) Vaihingen→Potsdam: We report experimental results on the Vaihingen → Potsdam cross-domain task in Table II. The source model is initially pretrained on 11 labeled Vaihingen images with approximately 2000 × 2000 pixels, and then adapted to the target domain in a source-free manner using 17 unlabeled Potsdam images with sizes of 6000 × 6000 pixels. This task poses a dual challenge of significant domain discrepancy and limited source supervision, allowing a comprehensive evaluation of SFDA adaptation performance under constrained conditions.

In the SFDA setting, our method achieves mIoU scores of 55.42% and 60.67% with the CNN-based and transformer-based architectures, respectively. Compared with source-trained baselines, the S^3T-SFDA improves the mIoU by

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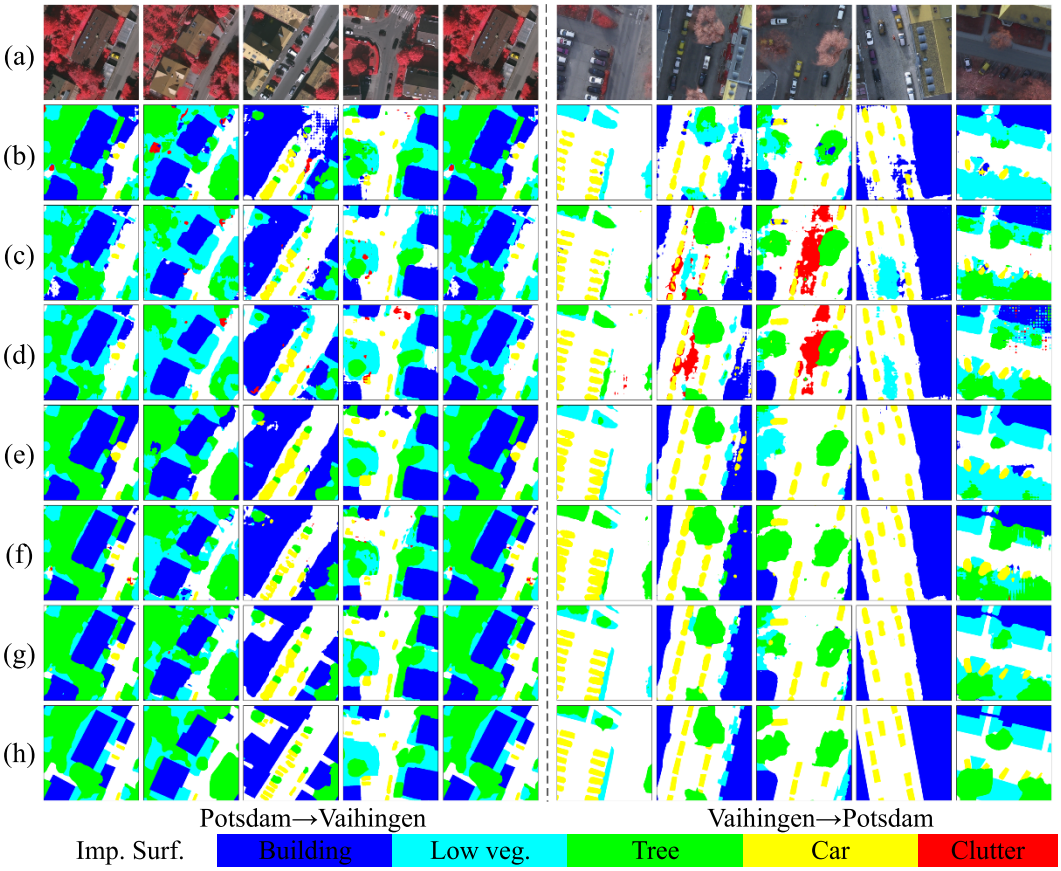


Fig. 7. Qualitative visualization comparison for the Potsdam→Vaihingen and Vaihingen→Potsdam tasks. (a) Target image. (b) Source-trained model. (c) HCL. (d) IAPC. (e) DT-ST. (f) S^{3}T-SFDA (DLv2). (g) S^{3}T-SFDA (SegF.). (h) Ground truth.

TABLE II

QUANTITATIVE COMPARISON RESULTS FOR THE VAIHINGEN → POTSDAM TASK. SF INDICATES WHETHER DOMAIN ADAPTATION WAS PERFORMED UNDER SOURCE-FREE CONDITIONS. PERFORMANCE VALIDATED ON THE POTSDAM TESTING SET

Methods	SF	Imp. Surf.		Building		Low veg.		Tree		Car		Clutter		mIoU	
		F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	F1	IoU	mFscore	mIoU
DeeplabV2	-	77.00	62.60	80.62	67.53	65.68	48.89	19.47	10.78	71.65	55.82	20.38	11.35	55.80	42.83
SegFormer	-	78.14	64.13	83.16	71.18	75.47	60.60	47.73	31.35	77.01	62.61	26.67	15.39	64.70	50.87
scaleDA[33]	✗	-	49.76	-	46.82	-	52.93	-	40.23	-	42.97	-	-	-	46.54
Zhang et al.[68]	✗	78.59	64.39	79.84	66.44	63.27	47.17	54.60	37.55	75.08	59.35	24.59	12.31	62.66	47.87
JDAF[69]	✗	80.74	67.70	86.59	76.36	67.72	51.19	53.17	36.21	77.47	63.22	23.17	13.10	64.81	51.30
DAFormer[13]	✗	83.49	71.66	90.83	83.21	67.76	51.24	78.8	65.01	70.28	52.07	35.44	21.54	71.10	57.45
ST-DASegNet[70]	✗	86.65	76.45	91.19	83.81	75.80	61.04	77.22	62.89	84.76	73.54	0.35	0.18	69.33	59.65
PFM-JONet[71]	✗	85.78	75.97	90.18	84.37	75.61	58.35	80.91	68.62	85.35	74.24	8.44	5.72	71.05	61.21
HCL[66]	✔	71.12	55.18	74.72	59.64	70.91	54.92	64.18	47.25	70.88	54.90	53.54	36.56	67.56	51.41
IAPC[56]	✔	71.40	55.52	75.13	.60.17	75.56	60.72	**67.17**	**50.57**	74.01	58.75	53.31	36.34	69.43	53.68
DT-ST[21]	✔	78.36	64.42	82.37	70.02	75.45	60.57	45.51	29.46	77.77	63.63	53.97	36.96	68.90	54.18
SND[20]	✔	73.77	58.44	77.15	62.80	76.13	61.46	61.13	44.01	75.00	60.00	53.16	36.20	69.39	53.82
S^{3}T-SFDA(DLv2)	✔	80.68	67.62	82.93	70.85	73.91	58.61	60.57	43.45	76.49	61.94	46.21	30.05	70.13	55.42
S^{3}T-SFDA(SegF.)	✔	**81.91**	**69.36**	**85.52**	**74.70**	**77.66**	**63.48**	61.13	44.02	**85.62**	**74.86**	**54.67**	**37.62**	**74.42**	**60.67**

12.59% and 9.8% and the mFscores by 14.33% and 9.72% for the two architectures, respectively. Under the same baseline and experimental settings, our method yields improvements of 0.74% and 1.6%, respectively, over the currently published SND and improvements of 1.23% and 1.24%, respectively, over DT-ST. Moreover, the SFDA methods, such as HCL and IAPC, perform better overall on minor categories such as cars and clutter. It is argued that the conventional UDA training process is more susceptible to overfitting because of the dominance of source domain-specific

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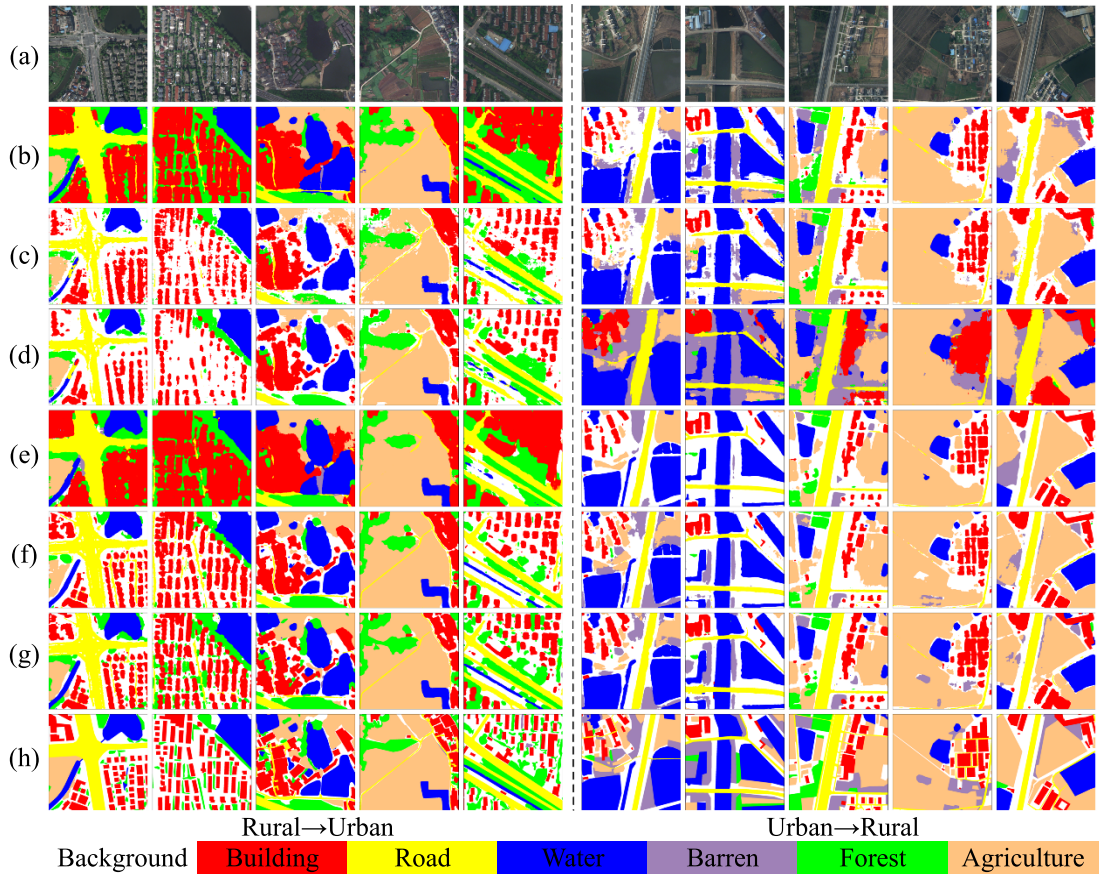


Fig. 8. Qualitative visualization comparison for the Rural→Urban task and Urban→Rural task. (a) Target image. (b) Source-trained model. (c) HCL. (d) IAPC. (e) DT-ST. (f) S^{3}T-SFDA (DLv2). (g) S^{3}T-SFDA (SegF.). (h) Ground truth.

knowledge, whereas the SFDA process focuses on exploiting the domain specificity of each category contained in the target domain, leading to more adequate adaptation to the target domain.

Fig. 7(b) presents the qualitative segmentation results of S^{3}T-SFDA and the comparison methods on the target domain. Existing approaches exhibit significant category confusion, particularly between spectrally and texturally similar classes such as buildings and impervious surfaces, or low vegetation and trees. S^{3}T-SFDA yields more accurate semantic delineation, benefiting from the ability of the proposed STC to model interclass semantic relations and enhance the discrimination between visually similar categories.

3) Cross Domain on LoveDA: To further evaluate the generalizability and stability of S^{3}T-SFDA in more complex RS cross-domain scenarios, we conduct experiments on the LoveDA dataset, covering two representative domain shifts:

Rural →Urban and Urban→Rural: The evaluation results are submitted to the official "LoveDA UDA" online benchmark for standardized assessment [73]. Compared with the ISPRS dataset, LoveDA poses significantly greater challenges due to increased scene complexity, larger urban–rural scale variation, and severe class imbalance, making source-free adaptation on this benchmark considerably more difficult.

The quantitative comparison results yielded by different methods for the Rural→Urban and Urban→Rural cross-domain tasks are demonstrated in Tables III and IV, where the proposed method significantly improves upon the segmentation performance of the baselines in the target domain. For the Rural→Urban cross-domain task, S^{3}T-SFDA improves the mIoU performance by 7.28% and 9.61% over that of the baselines based on CNNs and transformers, respectively. For the Urban→Rural cross-domain task, S^{3}T-SFDA improves the mIoU performance by 9.11% and 8.8% over those of the baselines, respectively. With DeepLabV2 as the segmentation architecture, S^{3}T-SFDA yields 2.64% and 3.3% improvements over the existing SOTA SFDA method in the two cross-domain tasks. In addition, with SegFormer as the segmentation architecture, the adaptation performance of the proposed method is substantially improved, with mIoU of 50.89% and 48.85%. Under the source-free setting, S^{3}T-SFDA outperforms most conventional UDA methods, such as DAFormer, with only a slight performance gap of 1.03%–1.23% compared to advanced approaches like ST-DASegNet. This gap arises primarily from the significant differences in class proportions and spatial distributions (e.g., bare soil and farmland) between urban and rural scenes. These findings highlight the remaining challenges in handling severe class imbalance and high cross-domain heterogeneity. Fig. 8 further visually illustrates the adaptation performance in two cross-domain scenarios. S^{3}T-SFDA delivers robust segmentation and strong generalization, particularly in densely built urban areas and in rural regions with complex distributions of water bodies. Overall, these results validate the effectiveness

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TABLE III

QUANTITATIVE COMPARISON RESULTS FOR THE RURAL → URBAN TASK. SF INDICATES WHETHER DOMAIN ADAPTATION WAS PERFORMED UNDER SOURCE-FREE CONDITIONS. TESTING PERFORMANCE IS SUBMITTED TO THE ONLINE PLATFORM FOR EVALUATION

Methods	SF	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU
DeeplabV2	-	41.05	37.58	24.40	75.65	15.83	33.40	31.43	37.05
SegFormer	-	45.66	47.24	41.12	82.53	14.47	27.34	30.61	41.28
CBST[46]	✗	48.37	46.10	35.79	80.05	19.18	29.69	30.05	41.32
IAST[47]	✗	48.57	31.51	28.73	86.01	20.29	31.77	36.50	40.48
DCA[31]	✗	45.82	49.60	51.65	80.88	16.70	42.93	36.92	46.36
DAFormer[13]	✗	50.94	56.66	62.83	89.41	11.99	45.81	25.26	48.99
ST-DASegNet[70]	✗	51.01	54.23	60.52	87.31	15.18	47.43	36.26	50.28
Geng et al.[78]	✗	37.05	61.79	56.15	60.54	44.28	50.45	53.19	51.92
HCL[66]	✔	38.60	39.86	27.30	79.69	10.88	41.32	33.26	38.70
IAPC[56]	✔	44.15	39.10	40.84	81.44	11.82	40.42	34.08	41.69
DT-ST[21]	✔	22.63	42.60	48.88	78.32	15.34	42.90	38.35	41.29
SND[20]	✔	25.46	42.94	45.55	77.67	16.07	41.01	43.34	41.72
Gao et al.[18]	✔	56.75	44.69	38.67	70.31	32.47	37.93	19.02	42.83
S^3T-SFDA(DLv2)	✔	41.70	48.57	51.20	77.62	14.98	30.59	**45.61**	44.33
S^3T-SFDA(SegF.)	✔	**46.15**	**56.50**	**60.12**	**84.78**	**16.18**	**48.29**	44.23	**50.89**

TABLE IV

QUANTITATIVE COMPARISON RESULTS FOR THE URBAN → RURAL TASK. SF INDICATES WHETHER DOMAIN ADAPTATION WAS PERFORMED UNDER SOURCE-FREE CONDITIONS. TESTING PERFORMANCE IS SUBMITTED TO THE ONLINE PLATFORM FOR EVALUATION

Methods	SF	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU
DeeplabV2	-	26.37	47.75	33.11	56.39	9.10	32.70	42.80	35.46
SegFormer	-	26.60	55.80	35.62	65.44	17.13	40.13	39.65	40.05
CBST[46]	✗	25.06	44.02	23.79	50.48	8.33	39.16	49.65	34.36
IAST[47]	✗	29.97	49.48	28.29	64.49	2.13	33.36	61.37	38.44
DCA[31]	✗	36.38	55.89	40.46	62.03	22.01	38.92	60.52	45.17
DAFormer[13]	✗	37.39	52.84	41.99	72.05	11.46	46.79	61.27	46.25
ST-DASegNet[70]	✗	36.78	59.83	43.77	73.83	19.38	49.96	67.01	50.08
Geng et al. [78]	✗	41.50	56.00	65.20	38.46	68.78	11.49	52.87	46.63
HCL[66]	✔	16.46	39.67	33.41	55.21	17.83	41.10	62.38	38.01
IAPC[56]	✔	28.55	56.22	36.77	63.47	11.22	38.45	51.04	40.82
DT-ST[21]	✔	29.54	56.13	36.12	64.16	12.32	37.71	52.88	41.27
SND[20]	✔	28.01	55.99	36.64	60.68	15.22	43.93	55.69	42.31
S^3T-SFDA(DLv2)	✔	30.21	58.77	34.25	60.05	**22.36**	43.68	62.66	44.57
S^3T-SFDA(SegF.)	✔	**37.92**	**60.59**	**45.68**	**72.76**	11.64	**46.10**	**67.24**	**48.85**

and robustness of S^3T-SFDA in adapting to diverse and challenging geographic environments.

E. Ablation Experiments

1) Effectiveness of the Components: To validate the effectiveness of each proposed component, we conduct ablation experiments on the Potsdam → Vaihingen cross-domain task using SegFormer and DeepLabV2 backbones, as presented in Table V. The source-only models achieve mIoU scores of 54.81% and 43.21%, respectively, revealing the substantial impact of domain shift. Introducing a basic ONST strategy yields only marginal improvements, reflecting the instability of pseudolabels generated under unsupervised conditions and their limited support for adaptation. Incorporating the DTU mechanism [21] improves mIoU to 61.14% and 49.22%, benefiting from momentum-guided teacher–student updates that stabilize model evolution. Building upon this, introducing the proposed SMVC further boosts performance by 2.54%, validating its effectiveness in mitigating semantic drift caused by local prediction errors through the multiview context fusion. Notably, the TDC mechanism further improves performance by 2.32% over ONST-DTU and shows complementary benefits when combined with SMVC, highlighting the role of temporal knowledge in enhancing training stability and alleviating knowledge forgetting. Furthermore, the STC strategy contributes a further 1.44% gain. Unlike SMVC and TDC, which focus on improving the pseudolabel consistency and reliability, STC imposes semantic constraints across spatial and temporal dimensions, effectively enhancing the discriminability of target-domain representations.

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TABLE V

ABLATION EXPERIMENTS FOR EACH COMPONENT OF S^{3}T-SFDA. Δ INDICATES MIOU PERFORMANCE GAINS COMPARED WITH SOURCE-TRAINED BASELINES

Network	ONST	DTU	SMVC	TDC	STC	SFDA	Δ
SegFormer						54.81	0
	✔					56.46	1.65
	✔	✔				61.14	6.33
	✔		✔			59.46	4.65
	✔	✔	✔			63.68	8.87
	✔	✔		✔		63.46	8.65
	✔	✔	✔	✔		66.34	11.53
	✔	✔	✔	✔	✔	67.78	12.97
DeepLabV2						43.21	0
	✔					44.94	1.73
	✔	✔				49.22	6.01
	✔	✔	✔	✔	✔	55.27	12.06

TABLE VI

EFFECTIVENESS ANALYSIS OF DIFFERENT SPATIAL CONSISTENCY STRATEGIES

Spatial Operation	mIoU
SFDA （w/o ROIAlign）	50.72
+Local Single View Consistency	50.92
+SMVC（w/o Fusion Align）	51.64
+SMVC	53.15

2) Effectiveness of Spatial Consistency: We evaluate the effectiveness of various spatial consistency modeling strategies for SFDA, as reported in Table VI. Specifically, ROIAlign is employed to generate local spatial features with different contextual views, including conventional single-view consistency [42], [57], multiview consistency, and the proposed SMVC with pseudolabel fusion and alignment. Under the baseline setting without any spatial consistency mechanism, the model achieves an mIoU of 50.72% on the target domain. Compared with the basic single-view strategy, introducing spatial multiview learning that captures both global and local context leads to superior performance, because single-view spatial consistency tends to ignore diverse contextual information. Notably, semantic alignment and fusion of predictions from different spatial views further improve mIoU by 2.43% over the baseline. This highlights that enforcing semantic consistency across spatial contexts effectively suppresses semantic drift caused by contextual variation, providing more stable and reliable pseudosupervision in source-free scenarios.

3) Analysis of Spatial Scales From Multiviews: In the proposed SMVC mechanism, the reconstructed spatial scale and the number of multiviews N equally impact adaptation performance. As shown in Table VII, we progressively adjust the original image resolution, the reconstructed scale, and the number of spatial views used for fusion. Observe that moderately increasing the number of views generally improves adaptation, benefiting from the complementary fusion of global and local spatial perspectives, which improves the consistency of pseudolabels and boosts generalization to the target domain. However, excessive view expansion (e.g., $N = 8$) slightly degrades performance, likely due to the inclusion of redundant information that introduces training bias. Meanwhile, more views also increase the computational overhead; we reduce the batch size to maintain manageable resource consumption. In terms of spatial scale, reconstructing images from 1024 × 1024 to a much lower resolution of 256 × 256 leads to significant performance drops, suggesting that aggressive downscaling causes the loss of critical spatial details, weakening feature representation. Overall, there exists a tradeoff among view number, spatial resolution, and computational cost. A configuration with a moderate reconstruction scale and a reasonable number of views ($N = 4$) achieves an effective tradeoff between accuracy and efficiency, facilitating more robust domain adaptation.

TABLE VII

SENSITIVITY ANALYSIS OF SPATIAL SCALES AND VIEW COUNTS

Origin size→Rebuild size		N	mIoU
1024×1024	512×512	1	50.92
1024×1024	512×512	2	52.46
1024×1024	512×512	4	53.15
1024×1024	256×256	8	51.26
512×512	256×256	2	52.38
512×512	256×256	4	53.17
512×512	256×256	8	52.53

TABLE VIII

ADAPTATION PERFORMANCE WITH TUNED HYPERPARAMETERS K AND M ACROSS DIFFERENT CROSS-DOMAIN SCENARIOS

cross domain scenarios	K=1, M=1	K=1, M=3	K=2, M=2	K=2, M=3	K=2, M=5	K=5, M=3	K=10, M=3	K=20, M=3
Potsdam→Vaihingen	64.15	66.34	65.18	67.78	65.24	64.85	62.32	62.10
Rural→Urban	50.40	52.45	52.17	53.44	51.84	52.41	51.21	49.12

4) Sensitivity of the TDC Hyperparameters: To investigate the sensitivity of the update interval K and the number M in the historical model queue, where K controls the frequency of teacher network updates in the queue, and M determines the total iteration interval of the current and oldest teacher networks. As presented in Table VIII, experiments are conducted on the Potsdam → Vaihingen and Rural → Urban cross-domain tasks. The hyperparameter K is fine-tuned within the interval (1, 20), and the hyperparameter M is tuned within the range {1, 2, 3, 5} to control the memory capacity. The best performance is observed when $K = 2$ and $M = 3$, with mIoU gains of 3.63% and 3.04% over standard momentum-based updating. This highlights the benefit of incorporating historical knowledge to stabilize the self-evolution process and mitigate performance degradation caused by knowledge forgetting. Nevertheless, with increasing K and M, the mIoU performance significantly decreases, indicating that the long-term inactivity of the historical model leads to outdated knowledge, which in turn exacerbates the adaptation performance degradation.

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TABLE IX
PERFORMANCE GAINS OF DIFFERENT SEGMENTATION NETWORKS APPLIED TO S^{3}T -SFDA

Backbone	Method	mIoU		
		Baseline	SFDA	Gain
ResNet101[75]	DeeplabV2[74]	43.21	55.27	12.06
EfficientNet-B2[79]	U-Net	48.41	56.10	7.69
MiT-B2[76]	SegFormer[76]	54.81	67.78	12.97
MiT-B4[76]	SegFormer[76]	55.48	69.03	13.55
ConvNeXt[80]	UperNet[81]	51.43	61.66	10.23
Swin-small[82]	UperNet[81]	54.30	65.04	10.74
Swin-base[82]	UperNet[81]	57.33	68.89	11.56

TABLE X
SENSITIVITY ANALYSIS OF THE LOSS WEIGHTS ON THE POTSDAM → VAIHINGEN CROSS-DOMAIN TASK

α	β	mIoU
0.05	0.5	67.16
0.1		67.78
0.2		67.32
0.5		66.48
1		66.87
0.1	0.3	66.91
	0.4	67.37
	0.5	67.78
	0.6	67.53
	0.7	67.12

5) Generalizability for Different Networks: To further evaluate the robustness of S^{3}T-SFDA under different backbone and segmentation approaches, as shown in Table IX, we select a variety of architectures based on CNNs and transformers as segmentation baselines. It can be seen that the segmentation models trained in the source domain all achieve stable performance improvement after adaptation, which validates the generality and stability in different network architectures. Nevertheless, compared with the CNN-based EfficientNet [79] and ConvNeXt [80] approaches, adopting the transformer-based MiT [76] and Swin [82] backbones results in superior adaptation capabilities in scenarios with source data shortages. Moreover, MiT and Swin backbones with different scales demonstrate that deeper and larger models achieve better baseline results, whereas the proposed SFDA approach remains stable with respect to its performance gains.

6) Sensitivity Analysis of Loss Weights: We conducted a sensitivity analysis of the loss weights on the Potsdam→Vaihingen task, as shown in Table X. Here, α and βrepresent the weights of the STC loss $\mathcal{L}_{\text{Con}}$ and information entropy loss $\mathcal{L}_E$ in (11), respectively, and their values were fine-tuned via cross validation. One can find that when the weight of the STC exceeds 0.2, significant performance fluctuations occur, whereas the information entropy loss is relatively less sensitive. The best performance is achieved when $\alpha = 0.1$ and $\beta = 0.5$, which are thus adopted as the default parameters in our experiments.

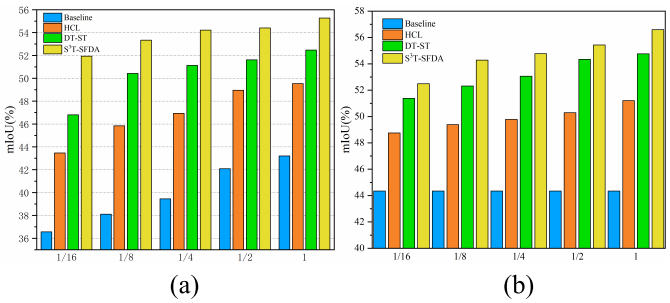


Fig. 9. Adaptation performance under varying data scales. (a) Validation with different source data proportions. (b) Validation with different target data proportions.

7) Stability With Limited Source Data: To assess the stability of the source-trained model adapted to the target domain under data constraints, we randomly select different proportions of the source and target data to participate in the training process. As shown in Fig. 9(a), we select different proportions of the source data for training on the Potsdam→Vaihingen cross-domain task to obtain source-trained weights, and adapt these weights to the target domain via different SFDA methods. To ensure fairness, all methods adopt DeepLabV2 as their baseline and follow the same experimental setup. Notably, the performance of the baseline in the target domain gradually decreases as the amount of source data decreases, and this poorer baseline further weakens the adaptation ability in the target domain. The proposed S^{3}T-SFDA shows better results than HCL and DT-ST for various amounts of data. With only 1/8 of the training data, S^{3}T-SFDA achieves comparable performance with that of the DT-ST method, sufficiently demonstrating its advantages in scenarios with limited data.

8) Stability With Limited Target Data: Fig. 9(b) shows the SFDA performance for different target data limitations in the Vaihingen→Potsdam cross-domain task. Each SFDA method performs domain adaptation with the same source-trained weights, but the proportion of target data involved in training varies across 1/16, 1/8, 1/4, 1/2, and all of the training data, where the 1/16 setting contains fewer than 200 training samples. Each SFDA method still exhibits strong generalizability under the target data constraints. The HCL method exhibits better stability despite its overall performance being inferior to those of DT-ST and S^{3}T-SFDA, while the proposed S^{3}T-SFDA shows significant advantages in terms of both its mIoU performance and its stability in target data-constrained scenarios. This is attributed to the spatial–temporal consistency learning strategy, which can effectively mine the semantic information of the target domain to improve the adaptability.

9) Analysis of Different Cross-Domain Directions: To investigate the impact of the cross-domain direction on SFDA performance, we conducted comparative experiments on Potsdam→Vaihingen and Vaihingen→Potsdam tasks, as shown in Table XI. In both cases, the source and target training sets were fixed to the same sizes (300 and 500 samples). It can be observed that Potsdam→Vaihingen achieves slightly better overall adaptation, likely due to the higher resolution and finer textures in Potsdam, which provide richer seman-

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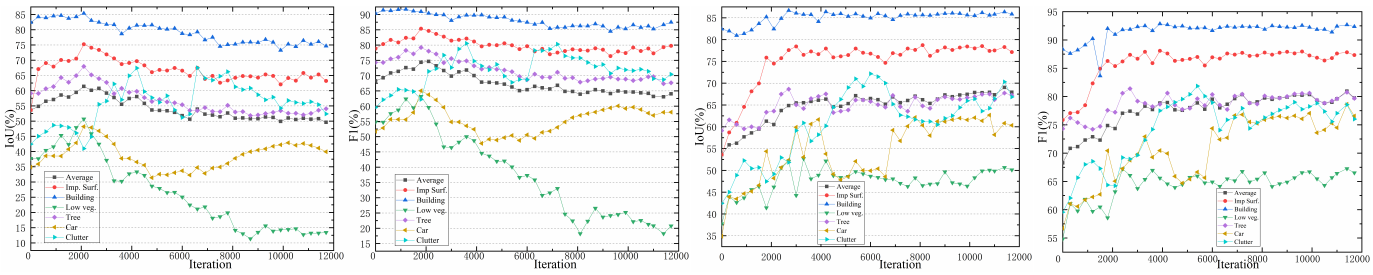


Fig. 10. Prediction curves of IoU and $F1$ scores during training for different SFDA methods. (a) IoU curve of conventional self-training. (b) $F1$ curve of conventional self-training. (c) IoU curve of S^{3}T-SFDA. (d) $F1$ curve of S^{3}T-SFDA.

TABLE XI

COMPARATIVE ANALYSIS OF ADAPTATION PERFORMANCE FOR DIFFERENT CROSS-DOMAIN DIRECTIONS

samples	cross-domain scenarios	baseline	S^{3}T-SFDA	Δ
300	Potsdam→Vaihingen	53.92	65.25	11.33
	Vaihingen→Potsdam	50.72	60.39	9.67
500	Potsdam→Vaihingen	55.02	65.78	10.76
	Vaihingen→Potsdam	50.81	60.76	9.95

TABLE XII

MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY OF SFDA METHODS

Methods	Params (M)	FLOPs (G)	Memory (MB)	Training efficiency (s)	Inference speed (fps)
DeeplabV2	42.94	179.54	4240	0.22	19.7
SegFormer	27.88	73.35	3522	0.18	19.3
IAPC	129.49	742.61	10478	0.73	19.7
SND	98.18	584.32	5253	0.42	19.7
S^{3}T-SFDA (DLv2)	144.57	667.55	5908	0.54	19.7
S^{3}T-SFDA (SegF.)	84.95	230.82	4536	0.43	19.3

tic and structural cues for cross-domain transfer. When the number of source samples is increased from 300 to 500, both directions exhibit consistent performance gains, confirming the positive impact of larger source sample scales on model generalization. Overall, S^{3}T-SFDA outperforms the baseline significantly in both cross-domain settings, with an average improvement of approximately 10%, demonstrating its strong robustness in cross-domain adaptation even under limited-sample conditions.

10) Predictive Stability of SFDA: To evaluate prediction stability during source-free iterative training, we plot per-class prediction accuracy curves for conventional SFDA methods and S^{3}T-SFDA, as shown in Fig. 10. For categories with clear structural boundaries, such as buildings and impervious surfaces, conventional SFDA achieves initial performance gains. However, as training continues, their predictions become increasingly unstable and gradually degrade. This suggests that the iterative process may suffer from noise accumulation, degrading the pseudolabel quality over time. The instability is more pronounced in semantically ambiguous classes such as low vegetation, indicating that conventional SFDA lacks robustness in uncertain or texture-varying regions. In contrast, S^{3}T-SFDA demonstrates significantly smoother and more stable prediction trends, with reduced fluctuations and superior accuracy across all classes. This fits our motivation: SMVC enforces spatial consistency by fusing multiview predictions, effectively reducing semantic uncertainty caused by blurry or shifting features, while TDC leverages historical knowledge to introduce temporal semantic stability, effectively suppressing training noise and knowledge forgetting.

11) Analysis of Computational Efficiency: To assess the computational efficiency of the proposed method, we compare various approaches on a single NVIDIA RTX 3090 GPU in terms of parameter counts, FLOPs, memory usage, training efficiency, and inference speed, as reported in Table XII. The IAPC method results in high memory consumption and training latency, primarily due to its computational complexity and the overhead introduced by its prototype storage mechanism. Our method incorporates the SMVC, TDC, and STC modules to achieve superior adaptation performance, using DeepLabV2 and SegFormer as backbone networks with parameter counts of 144.57M and 84.95M, respectively. Although these results are slightly higher than those of IAPC and SND, S^{3}T-SFDA maintains reasonable memory usage and training efficiency. This is attributed mainly to the fact that additional modules introduce computation primarily during the forward pass without significantly increasing the burden of gradient calculations in the backward pass. In addition, the memory overhead from storing historical outputs is effectively controlled, avoiding a significant memory burden. Overall, S^{3}T-SFDA achieves its performance gains at a reasonable computational cost. Notably, the additional components are only active during training and introduce no structural overhead during inference, ensuring that inference speed remains consistent with the original baseline.

12) Efficiency Analysis of Different Modules: To further assess the impact of each module on performance and overall efficiency, we conduct ablation experiments by incrementally introducing different modules under identical conditions. Table XIII reports the variations in training time and accuracy after adding each module, with performance results aligned with those in Table V. The DTU introduces an additional teacher network and update mechanism to enhance model stability. SMVC achieves significant performance gains with

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TABLE XIII
IMPACT ANALYSIS OF EACH COMPONENT MODULE ON PERFORMANCE AND OVERALL EFFICIENCY

Modules	Baseline	+DTU	+SMVC	+TDC	+STC
Training Time (h)	3.35	4.48	4.85	5.67	5.90
mIoU	56.46	61.14	63.68	66.34	67.78

minimal computational overhead through multiview alignment and prediction fusion. The primary computational burden of TDC derives from the introduction of historical teacher models and consistency training. STC requires only lightweight independent projection heads, with low-additional computational costs. Overall, although the combination of modules increases the training time from 3.35 to 5.90 h, the proposed design achieves a well-balanced tradeoff between accuracy and efficiency.

V. CONCLUSION

In this work, we propose an S^{3}T-SFDA approach, which focuses on the pseudolabel reliability and discriminative representation ability of the utilized model in the absence of any supervisory information, adapting a source-trained model to a target domain without access to the source data. To mitigate the prediction ambiguity across different local spatial perspectives, an SMVC mechanism is proposed to achieve improved pseudolabel stability in varying spatial contexts. To prevent the model from forgetting past knowledge and becoming overconfident in its current learning process due to the absence of supervision, a TDC mechanism is proposed to evaluate the prediction consistency between the current iteration and past moments within a certain dynamic iteration step. This mechanism also reweights the pseudolabels to regularize the evolution direction of the model. In addition, to alleviate the cognitive biases caused by domain shifts, the STC strategy is designed to improve the category-discriminative representations of different categories in the target domain. Experiments conducted on two domain-adaptive RS semantic segmentation benchmarks demonstrate the remarkable generalization performance of S^{3}T-SFDA in the target domain and its flexibility for adaptation to different advanced segmentation networks.

S^{3}T-SFDA is primarily tailored for source-free RS domain adaptation in the closed-set setting, where the source and target domains share identical categories. However, this assumption limits its applicability in more open scenarios. For instance, adapting models trained on generic datasets with different label systems to target domains remains an open challenge. In the future, we aim to extend our framework to more challenging scenarios, such as open-set adaptation with unseen target classes. In addition, we plan to investigate SFDA for RS foundation models and general-purpose segmentation models, which directly adapt the performance of generalized foundation models, such as the segment anything model (SAM), to specific RS scenarios without reliance on annotated data.

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